Exp no:-3
Date:-25/7/26

practical-3

AIM:

To Study the packet tracer tool Installation and user Interface Overview

To understand environment of CISCO PACKET TRACER to design simple network.

Introduction:

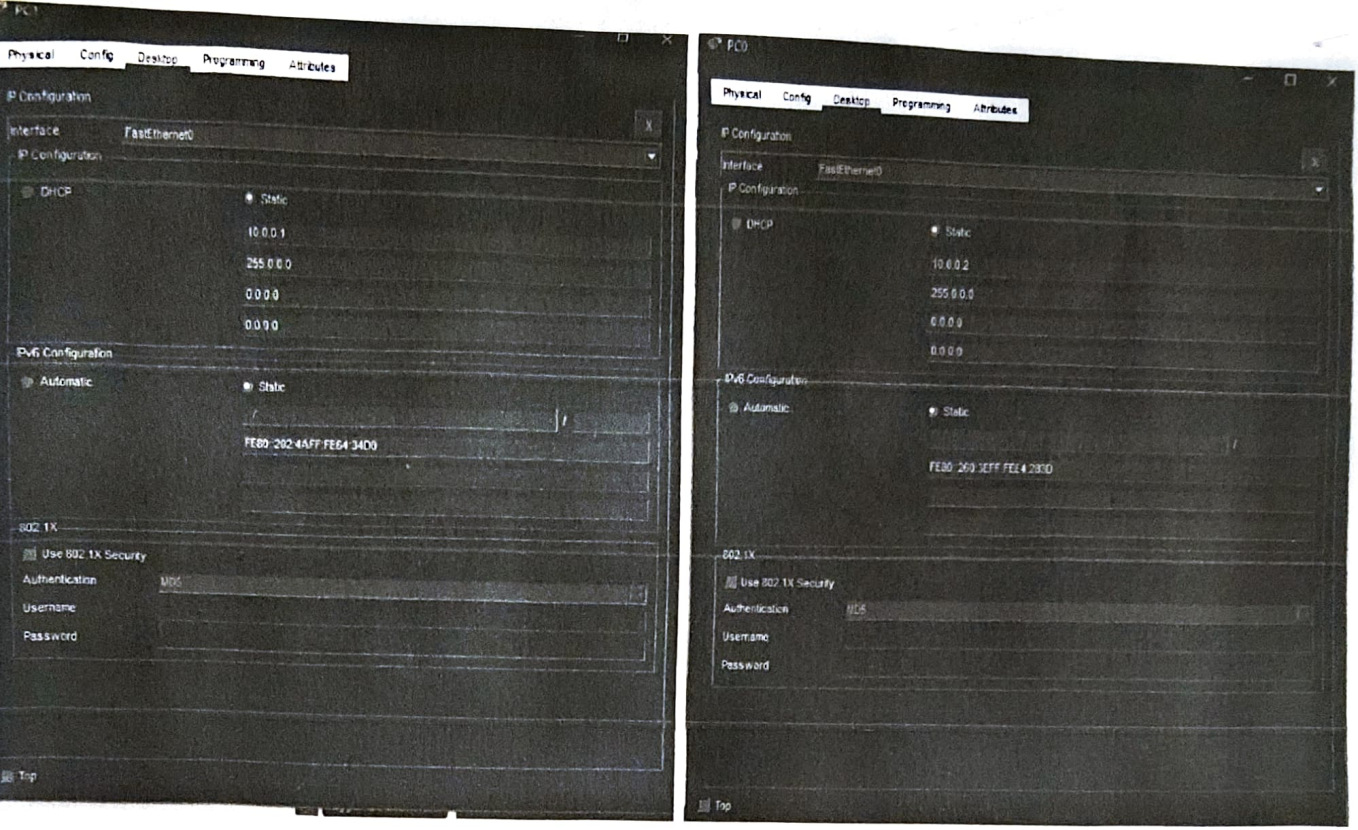
A Simulator as the name suggests, simulates network devices and its environment. Packet Tracer is an exciting network design, simulation and modelling tool.

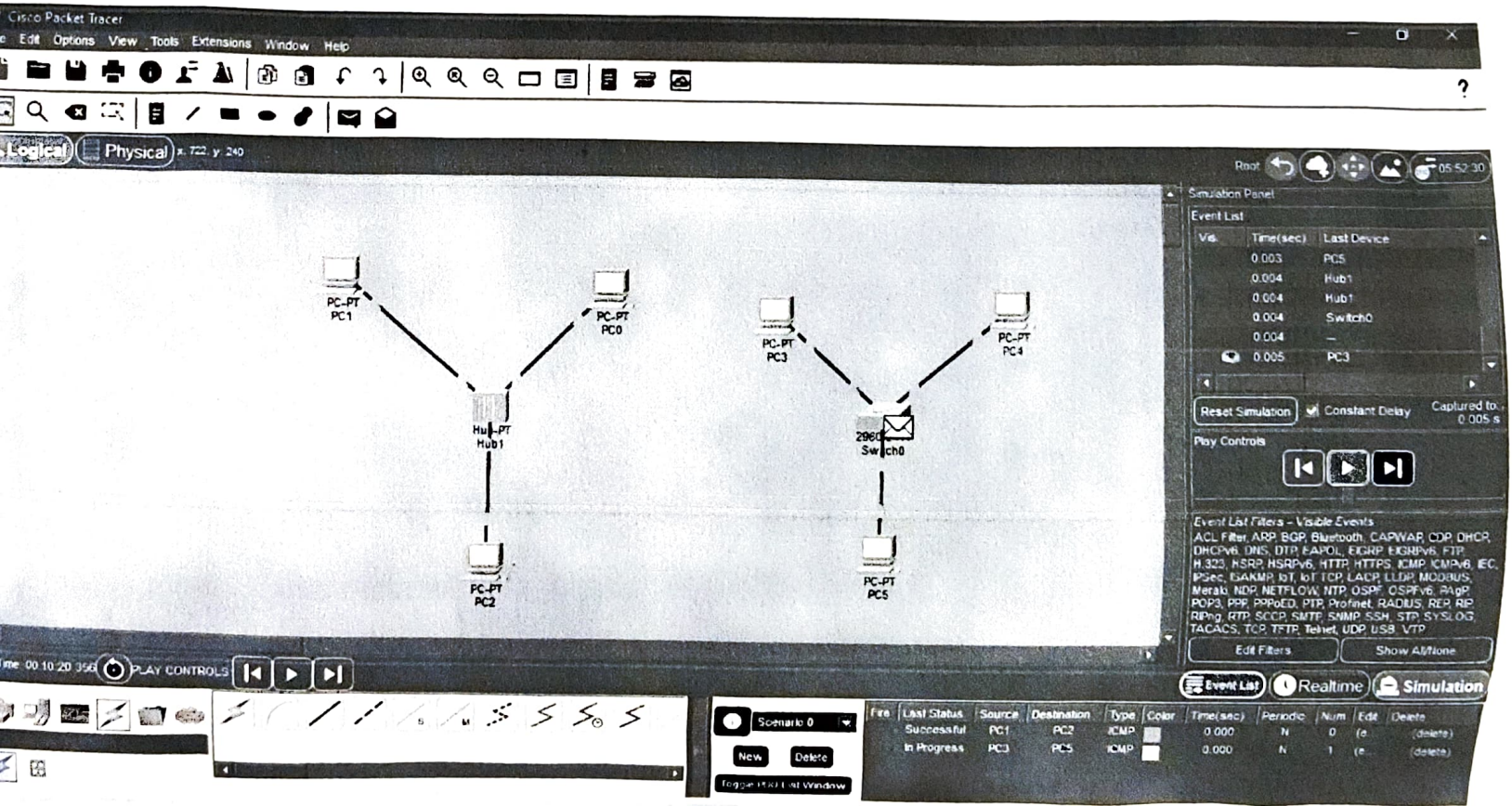
1. It allows you to model complex systems without the need for dedicated equipment.

2. It helps you to practice you to practice your network configuration and troubleshooting skills via computer or an Android or iOS based mobile device.

3. It is available for both the Linux and windows desktop environments.

4. Protocols in Packet Tracer are coded to work and behave in the same way as they would on real hardware.


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
10.0.0.1
255.0.0.0
0.0.0.0
0.0.0.0
IPv6 Configuration
Automatic
Static
FE80::202:4AFF:FE64:34D0
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
10.0.0.2
255.0.0.0
0.0.0.0
0.0.0.0
IPv6 Configuration
Automatic
Static
FE80::260:3EFF:FE64:2B0D
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 722, y: 240
Root
05:52:30
PC-PT PC1
PC-PT PC0
PC-PT PC3
PC-PT PC4
Hub-PT Hub1
2960-24TT Switch0
PC-PT PC2
PC-PT PC5
Simulation Panel
Event List
Vis.
Time(sec)
Last Device
0.003
PC5
0.004
Hub1
0.004
Hub1
0.004
Switch0
0.004
0.005
PC3
Reset Simulation
Constant Delay
Captured to:
0.005 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Time: 00:10:20.356
PLAY CONTROLS
Event List
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Successful
PC1
PC2
ICMP
0.000
N
0
(edit)
(delete)
In Progress
PC3
PC5
ICMP
0.000
N
1
(edit)
(delete)

User Interface Overview

The layout of packet Tracer is divided into several components. The components of the packet Tracer interface are as follows: match the numbers with explanations.

* Menu bar
* Main toolbar
* Logical / physical workspace tabs
* Workspace
* Common tools bar
* Real-time / Simulation tabs
* Network component box
* User-created packet box

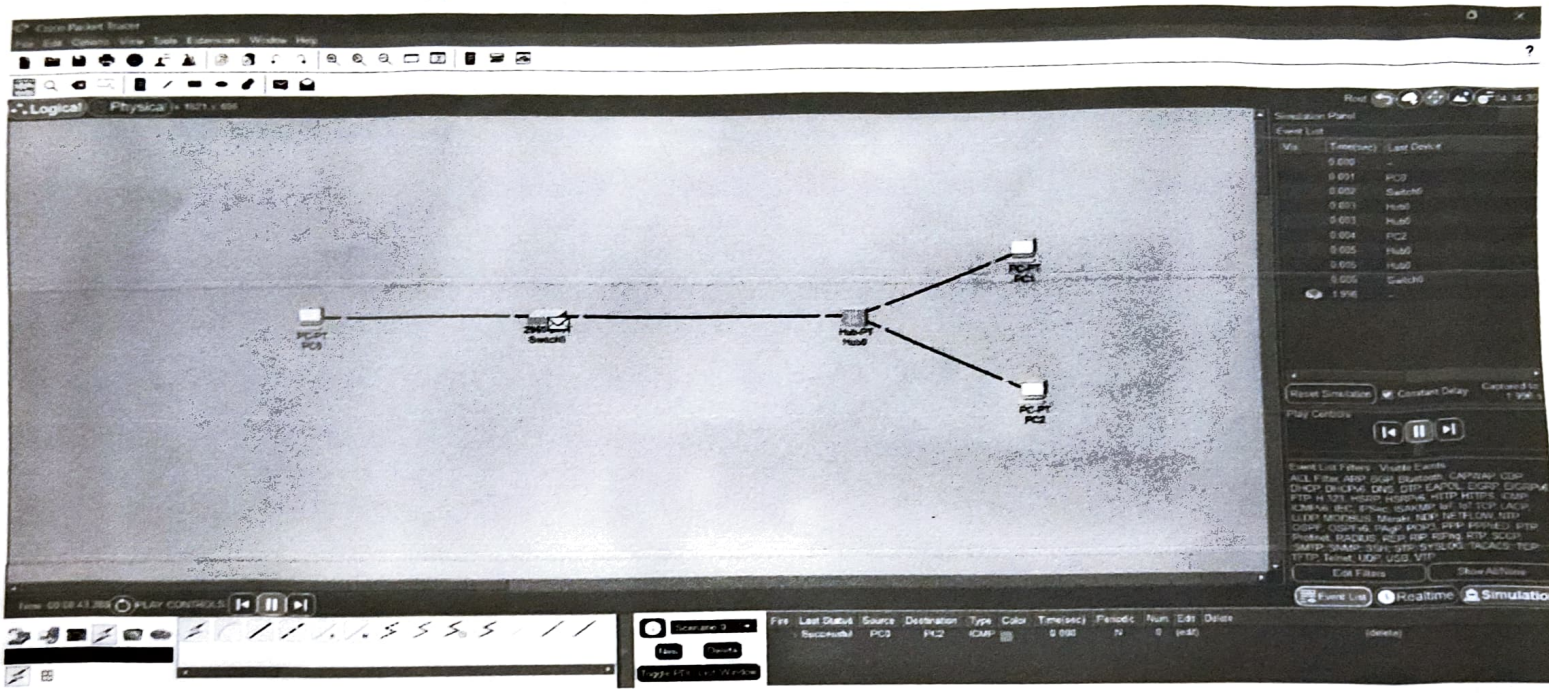
Analyse the behaviours of network device using CISCO PACKET TRACER simulator

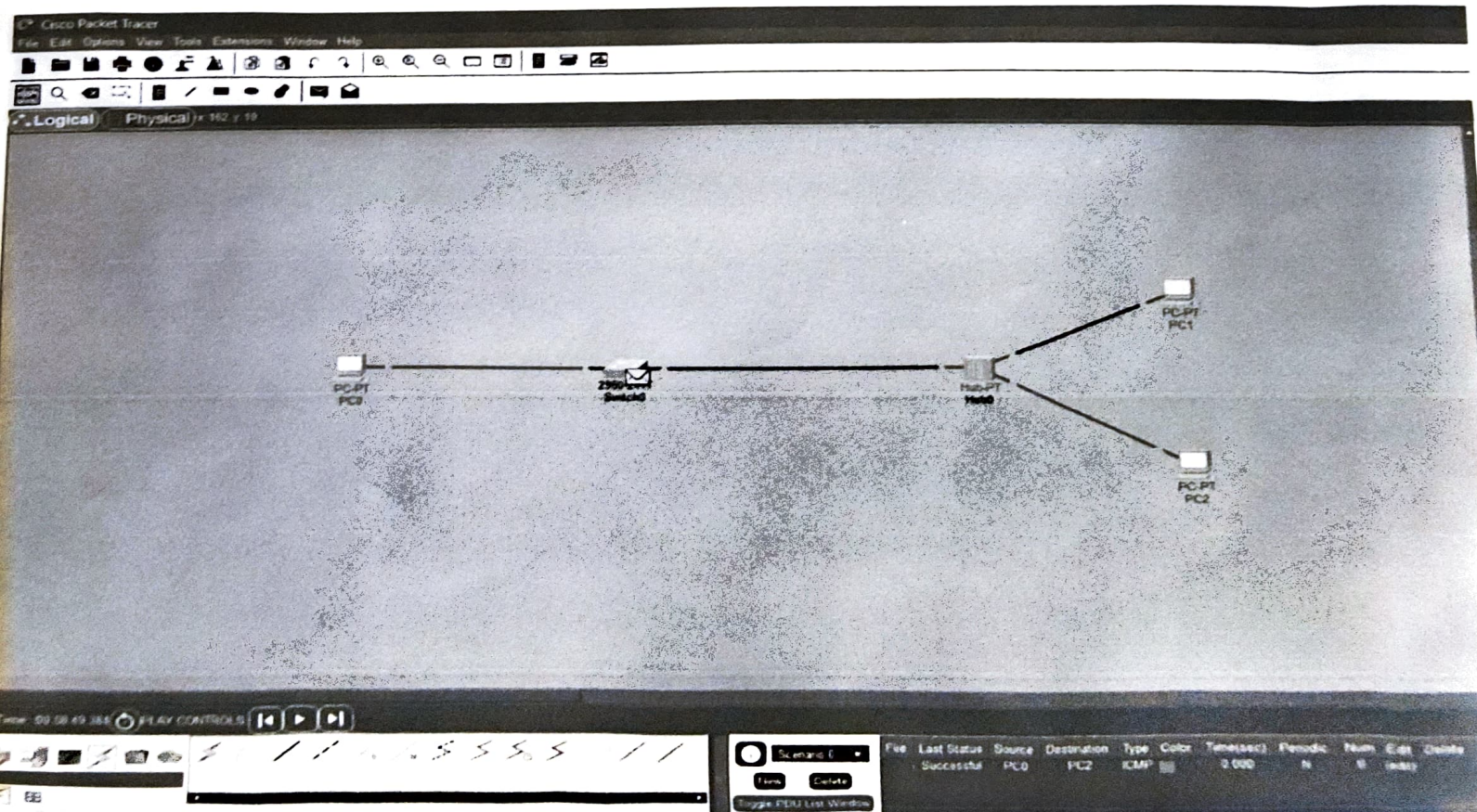
1. From the network component box. click and drag and drop the below components

   a. 4 generic PCs and one HUB
   b. 4 generic PCs and one switch

2. Click on connections:

   a. click on copper straight-Through cable
   b. Select one of the PC and connect it to


Cisco Packet Tracer
Logical
Physical
PC-PT PC0
2950-24 Switch0
Hub-PT Hub0
PC-PT PC1
PC-PT PC2
Realtime
Simulation


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
PC-PT PC0
2950-24 Switch0
Hub-PT Hub0
PC-PT PC1
PC-PT PC2
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC0 PC2 ICMP 0.000 N 0 (edit) (delete)

HUB using the cable. The link LED should glow in green, indicating that the link is up. Similarly connect remaining 3 PCs to the HUB.

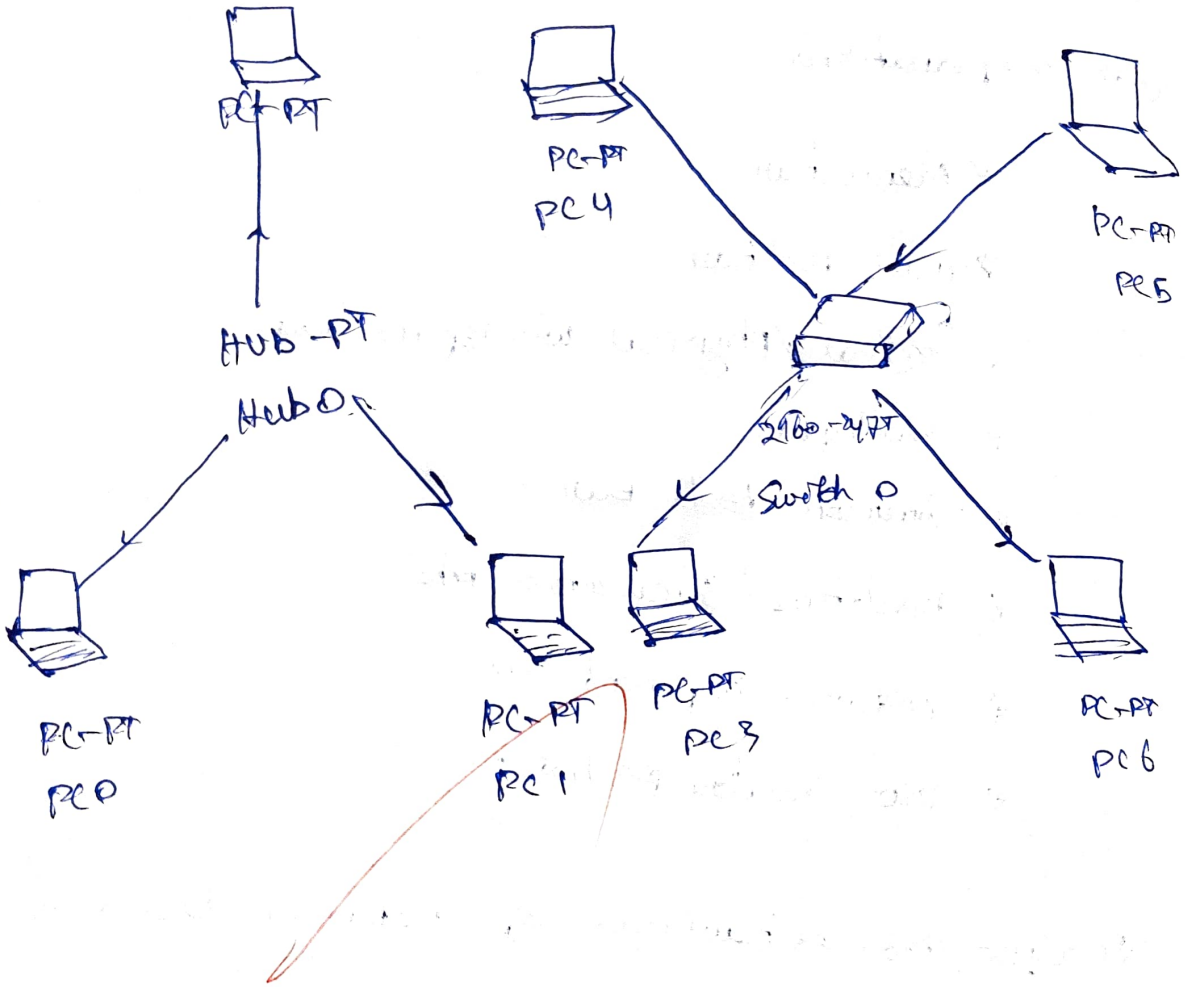
2. Similarly connect 4 PCs to the switch using copper straight-through cable.

3. click on the PCs connected to hub. go to the Desktop tab. click on IP configuration and enter an IP address and subnet mask. Here, the default gateway and DNS server information is not needed as there are only two end devices in the network.

4. Observe the flow of PDU from source PC to destination PC by selecting the Realtime mode of simulation.

5. Repeat Step # 3 to step #4 for PCs connected to the switch.

6. Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviour of switch and HUB.


PC-PT
PC4
PC-PT
PC5
HUB-PT
Hub 0
2960-24TT
Switch 0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC3
PC-PT
PC6

Result:

Thus the study of packet tracer tool Installation and user Interface was done successfully